



Ankan Sarkar
Computer Science & Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 15/10/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	Yogiraj Public School	2021	87.20%
Matriculation	CBSE	Bharatiya Vidya Bhavan	2019	96.80%

SCHOLASTIC ACHIEVEMENTS

- Secured an **All India Rank** of **67** in **JEE Advanced** among over 150 thousand registered candidates. [2021]
- Acquired an **All India Rank** of **92** in **JEE Mains** for which more than a million candidates appeared. [2021]
- Obtained **All India Rank 88** in **KVPY SX** stream among 50K students and was offered **IISC** fellowship. [2021]
- Secured an **All India Rank** of **684** in **KVPY SA** stream for which around 50 thousand students appeared. [2020]
- Cleared both stages of **National Talent Search Examination** and getting annual scholarship from Indian Govt. [2019]
- Ranked among **National Top 1%** in **National Standard Examination in Physics (NSEP)** for **two** years. [2019, 21]
- Received a **National Top 1%** certificate in **National Standard Examination in Chemistry (NSEC)**. [2021]
- Obtained certificate for **State Top 1%** in **National Standard Examination in Astronomy (NSEA)**. [2019]
- Bagged a prestigious **bronze medal** in **South East Asian Mathematics Olympiad (SEAMO)**. [2018]

WORK EXPERIENCE

Fraud Detection Server Development

[Summer 2024]

Associate

Samsung Electronics Co. Ltd.

- Developed the admin server of Samsung account's fraud detection system using **Spring Boot** and **DynamoDB**.
- Provided an integrated management and easy distribution of fraud blocking rules using **Drools** as a rule engine.
- Implemented **dynamic class loading** from schema files allowing sharing of resources from inspector to admin server.

AI-Powered Video Shorts Generator

[Summer 2023]

Software Trainee

Proshort India Pvt. Ltd.

- Developed a web application using **AI algorithms** to generate shorts from long videos using **uvicorn** & **fastapi**.
- Videos can be **uploaded from device** or via **URL** of either drive or YouTube allowing for a wide range of input.
- Users can **edit content** by adding, deleting, and rearranging sentences along with access to complete transcript.
- Implemented **user library** functionality for storing and downloading videos, ensuring enhanced flexibility for users.

KEY PROJECTS

Improvement of xv6 OS

[Autumn 2023]

Guide: Prof. Purushottam Kulkarni | Course Project: Operating Systems

IIT Bombay

- Implemented handling of **page faults** and executed **lazy allocation** of physical pages by modifying xv6 source code.
- Coded an **efficient process scheduler** using priorities and time slices along with detailed analysis of race conditions.
- Incorporated **threading processes** and implemented concurrency control mechanisms like **semaphores & barriers**.
- Looked at file system implementations to **encode protected files** and establish **symbolic links** among them.

Compiler for C-like Language

[Spring 2024]

Guide: Prof. Uday Khedker | Course Project: Implementation of Programming Languages

IIT Bombay

- Built a compiler for C from scratch using programs like **Flex & Bison** to perform lexical & syntactic analysis.
- Wrote clean modular code to generate intermediate representations like parse tree, abstract syntax tree, three address code & register transfer language to facilitate the production of assembly code with respect to **MIPS architecture**.

Service Rating Website

[Autumn 2022]

Guide: Prof. Kavi Arya | Course Project: Software Systems Lab

IIT Bombay

- Developed a **web app** enabling users to find reviews for restaurants, hotels, and theatres based on their location.
- Implemented a comprehensive **search feature** allowing users to explore a range of places within a **specified radius**.
- Enabled users to **add, edit, and delete reviews**, with options to sort places by average rating or distance.

Cache & Prefetcher Optimization

[Spring 2023]

Guide: Prof. Biswabandan Panda | Course Project: Digital Logic Design & Computer Architecture

IIT Bombay

- Implemented cache replacement policies (**Hawkeye**, **LIP**, **EAF**) and analyzed their impact on LLC miss rate, IPC.
- Developed and implemented the **best offset prefetcher** to enhance upon the next line and IP stride prefetchers.
- Experimented by modifying **cache exclusivity**, **number of sets** and **associativity** to evaluate cache efficiency.

Stock Trading Simulator

CS50x - Intro to Computer Science

[Spring 2022]

Harvard University

- Developed a Trading Simulator involving **Flask** that allows users to buy and sell stocks at **real-time share prices**.
- Implemented **registration system** and **transaction history** features, displaying exact date and time of dealings.

Railway Enquiry Portal

[Autumn 2022]

Guide: Prof. Supratik Chakraborty | Course Project: Data Structures and Algorithms Lab

IIT Bombay

- Implemented a railway portal in C++ with features for **train search, route listing, and review management**.
- Utilized a range of data structures including **AVL trees, hash tables, heaps, priority queues, tries and graphs**.
- Integrated functionalities for **user-generated reviews**, enabling search by keywords or by threshold rating values.

Image Inpainting

[Spring 2024]

Guide: Prof. Suyash Awate | Course Project: Medical Image Computing

IIT Bombay

- Provided a **detailed comparison** of various image inpainting methods by training on both faces and medical images.
- Coded various models involving **classical methods, autoencoders, variational autoencoders & DC GANs**.
- Output images were evaluated under **MSE, PSNR & SSIM** indices and the presence/absence of **poisson blending**.

CS50 AI

[Winter 2022]

Intro to Artificial Intelligence with Python

Harvard University

- Coded algorithms for **tic-tac-toe** solver, **logical puzzle** solver, **bot building** for Q/A and parsing sentences, **minesweeper** solver, **markov chains**, **crossword** solver, **nim** solver and **CNN** model as part of a **7 week** course.

OTHER PROJECTS

Image Processing

[Autumn 2022]

Guide: Prof. Suyash Awate | Course Project: Data Analysis & Interpretation

IIT Bombay

- Applied **PCA** to approximate linear relationship between variables and to **reduce dimension** of image matrices.
- Computed **principal modes** of variation to identify closest image representation as a combination of the eigenvectors.

Neural Style Transfer

[Autumn 2023]

Guide: Prof. Preethi Jyothi | Course Project: AI & ML

IIT Bombay

- Recreated a 2019 paper to implement neural style transfer using **mask-guided autoencoder**, utilizing a generative adversarial network with **perceptual and style losses** by training on a combination of content & style images.

File Transfer Protocol

[Spring 2023]

Guide: Prof. Bhaskaran Raman | Course Project: Computer Networks

IIT Bombay

- Developed a **multi client** file sharing protocol using **socket programming** allowing **two-way** sending of files.
- Implemented **flow control** to make sure that the sender side does not overwhelm the receiver side's resources.

Virtual Lunar Lander

[Autumn 2022]

Machine Learning Specialization

Stanford University

- Built a **deep Q-learning** neural network involving **experience replay** to simulate a **virtual lunar lander**.
- Involved the understanding of **Bellman equations, state-action value function & continuous state spaces**.

Game Development

[Winter 2022]

Self Project

IIT Bombay

- Developed **cross-platform compatible** games in **C++ & lua** using the simplecpp library and the love framework.
- Implemented popular games like Pong, Breakout, Snake Xenia, Flappy Bird, Bubble Trouble with **authentic UX**.

TECHNICAL SKILLS

- **Programming** - C, C++, Python, Java, Prolog, JavaScript, Bash, Awk, Sed, Lua, Matlab, MySQL, L^AT_EX, CMake
- **Development** - Spring Boot, Flask, Git, OpenCV, Django, Love, Uvicorn, Fastapi, Doxygen, Sphinx, NodeJS
- **Data Science/ML** - Keras, TensorFlow, Matplotlib, NumPy, Scikit-Learn, Pandas, OpenAI, Whisper, Spacy, NLTK

COURSES UNDERTAKEN

Operating Systems[†], Computer Architecture[†], Data Structures & Algorithms[†], Computer Networks[†], Applied Algorithms, Computer Graphics, Data Analysis & Interpretation, Software Systems Lab, Design & Analysis of Algorithms, Logic for CS, Automata Theory, Discrete Structures, AI & ML[†], Implementation of Programming Languages[†], Database Implementation Systems[†], Automatic Speech Recognition, Medical Image Computing, Numerical Analysis, Machine Learning for Remote Sensing, Decision Analysis & Game Theory, Abstractions & Paradigms in Programming[†], Computer Programming & Utilization, Matrix Computations, Quantum Information & Computing*, Systems Theory*

* To be completed by November'24

[†] Theory + Lab

EXTRA-CURRICULAR ACTIVITIES

- Secured **1st** position, represented my hostel in a team of 3 for a **Competitive Programming** Tournament. [2022]
- Secured **2nd** position in **Bodybuilding** Championship by Weightlifting Cell, IITB in my weight category. [2022]
- Awarded a **special mention** certificate for representing hostel in various tech related General Championships. [2023]
- Achieved **top 30** in MTRP and was invited twice to Indian Statistical Institute, Kolkata for 2-day camp. [2018, 19]